

YIFAN JING

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EDUCATION

University of Michigan, BS in Data Science

Expected 2027

Research Interests: Reinforcement Learning, Vision Language Models, AI Evaluation

Publication/Preprint/Manuscript

Xueyang Yu, **Yifan Jing**, Taiming Lu, Linrong Cai, Zhipeng Cai, Yuniang Xiong, Gabriel Herbert Sarch, and Zhuang Liu. *Can Image Generation Benefit Visual Reasoning?* **Manuscript in preparation. Under Review: ICLR 2027.**

Yifan Jing, Qining Zhang, Lei Ying. HintFlow: Small Models Learn to Steer Frozen Large Models through Learnable Hint Policies **Manuscript in preparation. Target: ICML 2027.**

Yifan Jing, Emma Yicheng Lou, Shaotian Sun, and Yuxuan Wu. *Computing Tools for Translation-Invariant Total Orders.* **arXiv preprint**, 2026. [arXiv:2607.11709]. **Target: FPSAC 2027.**

RESEARCH EXPERIENCE

Zlab, Princeton University

July 2026-Current

Role: Research intern under Prof. Zhuang Liu's supervision

- Investigating generation-augmented multimodal reasoning, designing systematic evaluations and controlled ablations to understand when and why generated visual thoughts help or hurt VLM reasoning across models and benchmarks.
- Developing and leading a new research direction on self-evolving AI / AI scientists, exploring agents that autonomously propose, evaluate, and iteratively improve reasoning strategies and solutions.

SURE 2026, University of Michigan

May 2026- Aug 2026

Role: Undergrad Researcher under Prof. Lei Ying's supervision

- Developed **HintFlow**, a framework that trains a 4B language-model policy to guide frozen 14B–20B reasoning models, via selective SFT and a modified GRPO objective with virtual-positive normalization for all-failed groups.
- Built the end-to-end training and inference pipeline with Qwen3-4B, Qwen3-14B, GPT-OSS-20B, LoRA, and vLLM;

Contact Sensing Hexapod, BIRDS Lab, University of Michigan

Jan 2026 - Current

(Contribute simulation part of this work, <https://arxiv.org/abs/2603.09147>)

Role: Undergraduate Researcher under Prof. Shai Revzen's supervision

- Developed and conducted large-scale simulation experiments on Mujoco for a multi-legged robotic locomotion system, evaluating performance across varying leg counts (3 segments-64 segments) and terrain conditions.
- Implemented and tested modular control policies in simulation, validating stability and adaptability on irregular terrains.
- Investigating compact neural policies with only thousands of parameters to optimize finite-state-machine control while preserving computational efficiency and interpretability.

Hybrid Transformer Analysis in Induction Head Problem, University of Michigan

Sep 2025 - Dec 2025

(details in this Github Repo <https://github.com/Ivann1242/Analysis-AR-problem>)

Project

- Studied generalization behavior of sequence models through the Associative Recall (AR) task, comparing Transformer, Mamba, and hybrid architectures.
- Proposed and evaluated hybrid architectures combining state-space models with attention mechanisms, demonstrating improved long-range generalization performance.
- Designed controlled experiments to analyze convergence and generalization dynamics across model families.

TITO (Translation-Invariant Total Orders), University of Michigan

Sep 2025 - April 2026

(details in 2025 Fall: <https://lsa.umich.edu/math/undergraduates/research-and-career-opportunities/research/LoGM/projects.html>)

Role: Researcher under Dr. Grant Barkley's advise

- Developed a **new algorithmic approach** for Translation-Invariant Total Orders (TITO), formalizing order-comparability through inversion sets and proving its correctness.
- Translated the theoretical model into a **computable structure**, enabling efficient algorithms for sequence comparison and ranking, with potential applications in sorting and discrete optimization.

- Provide a python package for TITO weak order comparison, JOIN computation, Normalization, visualization

TEACHING EXPERIENCE

University of Pittsburgh, Pittsburgh, Pennsylvania

August 2024 - Dec 2024

Undergraduate Teaching Assistant, Intermediate Programing

- Lectured undergraduate class of 125+ students; taught Java programming techniques in weekly office hours, requested recitations, and exam preparation sessions.

RELEVANT EXPERIENCE/HONOR

Tartanhacks <https://github.com/danielryang/spaceoverflow>

- Led 4 undergraduate students and won Top 5 Grant Over 300 submitted projects/700 participants in Tartanhacks 2026 hosted by CMU.

CAST-USA 33rd Innovation Summit

- Got Finalist reward at this most significant Chinese Innovation Competition in the USA, award presented by Prof. Avi Wigderson, 2023 ACM A.M. Turing Award Laureate.
- Presented CEEG AI Agent Business Plan in front of 300 audience.

SKILLS

Languages: Python(Jax, torch, numpy), C++, Java, HTML, javascript

Engineer: MuJoCo, Linux

Tools: GitHub Actions, Jupyter Notebook, Colab, Google Cloud Platform, LaTeX, Coding Agent

Soft Skills: VC-facing communication